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GMRIT/ADM/F-03
REV.: 03

# COURSE HANDOUT

## B. Tech – 5th Semester

Course Title : **Artificial Intelligence and Machine Learning Lab** Date: 27.07.2024

Course Code : 21CS507 Academic Year: 2024-2025

Course Structure : 0-0-3-1.5

**Course coordinator : Dr. R. Cristin**

**Instructor(s) : Dr. P. Kanchanamala, Dr. D. Sowjanya**

**Course Description:** The Artificial Intelligence and Machine Learning Lab course is a hands-on companion to the theoretical concepts covered in the AIML theory course. This lab-based course is designed to provide students with practical experience in implementing, experimenting with, and analyzing various rule-based, classification, clustering algorithms and techniques. Through a series of guided exercises and augmented project students will gain the skills and confidence needed to apply AI and ML methods to real-world problems.

**Scope and Course Objective:** The scope of an Artificial Intelligence and Machine Learninglab course is to provide hands-on experience and practical skills to complement the theoretical knowledge gained in the classroom. Lab courses play a crucial role in helping students apply AI and ML concepts to real-world problems, enhance their technical abilities, and prepare them for careers in various industries. AIML lab course is focused on providing students with practical skills, hands-on experience, and a deeper understanding of implementing AI and ML techniques in real-world scenarios. This lab course aims to bridge the gap between theoretical knowledge and practical application, enabling students to effectively utilize AI and ML tools and methods.

**Lab Manual**

1. Artificial Intelligence and Machine Learning Laboratory – Department of CSE, GMRIT, Rajam

**Text Book (s)**

1. Russel Norvig, Artificial Intelligence A modern Approach, 3rd Edition, Pearson Education 2010
2. Stephen Marsland- Machine Learning – An Algorithmic Perspective – Second Edition – Chap Man & Hall CRC Press, 2015

**Reference (s)**

1. E. Rich K. Knight and B.Nair – Artificial Intelligence– Third Edition – Tata McGraw Hill, 2017
2. Tom M. Mitchell, —Machine Learning, McGraw-Hill Education (India) Private Limited, 2013.

**Course Outcomes**

At the end of the course, the students will be able to:

1. Design and develop problem statements for solving real time problems using expert systems
2. Outline the role of data repositories and various data preprocessing tools
3. Illustrate the use of consistent data in implementing machine learning algorithms
4. Distinguish valid datasets for the machine learning algorithms
5. Identify and develop various machine learning algorithms suitable to the problem statement
6. Demonstrate and resolve complex clusters using dimensionality reduction

**CO–PO Mapping**

| CO | $PO_4$ | $PO_5$ | $PO_8$ |
|---|---|---|---|
| 1 | 3 | 2 | 2 |
| 2 | 3 | 3 | 2 |
| 3 | 3 | 1 | 3 |
| 4 | 2 | 2 | 2 |
| 5 | 2 | 2 | 3 |
| 6 | 1 | 2 | 3 |

3–Strongly linked | 2–Moderately linked | 1–Weakly linked

**List of Experiments**

**Expert System**

Week 1: Develop a Rule Based system (Consider any Problem from the Annexure any Real Time)

**Data Pre-Processing and Data Pre-Processing Tools**

Week 2: Importing Libraries, Importing Datasets, Missing Data, Encoding Dependent Variable,

Week 3: Splitting Datasets to Training Dataset and Test Dataset, Feature Scaling

**Classification:**

Week 4: K-Nearest Neighbor(K-NN) Classifier

Week 5: Support Vector Machine Classification

Week 6: Naïve Bayes Classifier

Week 7: Decision Tree Classification

**Regression**

Week 8: Simple Linear Regression

Week 9: Multiple Linear Regressions

Week 10: Polynomial Regression

Week 11: Logistic Regression

**Clustering**

Week 12: Association Rule Learning

Week 13: K Means Clustering

Week 14: Hierarchical Clustering

**Dimensionality Reduction**

Week 15: Principal Component Analysis

Week 16: Linear Discriminate Analysis

**Course Plan**

| Lab Session No. | Learning objectives | Topic(s) to be covered |
|---|---|---|
| 1. | Introduction | Installation of Python Tool, Anaconda navigator and Related Tools. |
| 2. | Develop an Expert System | **Week 1** To develop a Rule Based system (Consider any Problem from the Annexure any Real Time) |
| 3. | To Apply the Data Pre-Processing and Data Pre-Processing Tools | **Week 2** Importing Libraries, Importing Datasets, Missing Data, Encoding Dependent Variable |
| 4. | | **Week 3** Splitting Datasets to Training Dataset and Test Dataset, Feature Scaling |
| 5. | To Explore the Classification Algorithms. | **Week 4** K-Nearest Neighbours |
| 6. | | **Week 6** Support Vector Machines |
| 7. | | **Week 7** Naïve Bayes |
| 8. | | **Week 8** Decision Tree Classification |
| 9. | To Apply the Various Regression Techniques. | **Week 9** Simple Linear Regression |
| 10. | | **Week 10** Multiple Linear Regressions |
| 11. | | **Week 11** Polynomial Regression |
| 12. | | **Week 12** Logistic Regression |
| 13. | To Build a Model for Clustering Techniques | **Week 13** Association Rule Learning |
| 14. | | **Week 14** K Means Clustering |
| 15. | | **Week 15** Hierarchical Clustering |
| 16. | To Analyze the Dimensionality Reduction Concepts. | **Week 16** Principal Component Analysis |
| 17. | | **Week 17** Linear Discriminate Analysis |
| 18. | Internal Lab Exam | |

**Annexure:**

1. Banking System
2. Weather System
3. Medical System
4. Gaming System
5. Dressing System
6. Disease Diagnose or Prediction System

**Augmented Experiments:** (Students shall opt any one of the Augmented Experiments in addition to the regular experiments)

1. Rule Based Expert System: Design, Suiting and Shirting for an event/occasion
2. Gaming: Chess, Water Jug Problem, 8 Queens Problem
3. Real Time: Traffic Lights
4. Real Time: Banking Transactions (Only Debit and Credit)
5. Mathematical Approach for Theorem Proving
6. Financial Approach for Loans via Cash / Vehicle / Gold
7. Medical Expert Systems
8. Disease Classification Strategies
9. Disease Prediction Strategies
10. Work on a Real Dataset

**Evaluation Scheme:**

| Component | Duration (minutes) | Marks | % *Weightage* | Date & Time | Venue |
|---|---|---|---|---|---|
| Day to Day Performance | 120 | 10+5 | 20 | 29.07.2024 to 20.11.2024 | 5th Block |
| Internal Lab Test | 180 | 5 | 6.67 | 11.11.2024 to 20.12.2024 | |
| Augmented Experiment | --- | 5 | 6.66 | 11.11.2024 to 20.12.2024 | |
| Semester End Exams | 180 | 50 | 66.6 | 28.11.2024 to 14.12.2024 | |

**Chamber Consultation Hour**: Every Saturday 2.00 PM to 3.00 PM

**Venue:** CSE Staff Room

**Notices:** CSE Notice Board

| **Signature of the Instructors** | **Course-Coordinator** | **HOD-CSE** |
|---|---|---|
| a. Dr. P. Kanchanamala | Dr. R. Cristin | Dr. A. V. Ramana |
| b. Dr. D. Sowjanya | | |